

5COSC024W Server-Side Web Development 2021-2022 TUTORIAL 02 – INDEX – **MODEL ANSWER**

TASK 01| Create a Product table in the MySQL database

- 1) Logon to your MySQL account and access your database called **yourWestminsterID_0** (See “Software Guide” on Blackboard).
- 2) Create a **Product** table in MySQL containing 8 columns as shown on the logical ERD and according to the specifications below.

Column	Type	Length/values	Default	Default value	Null	Index	Auto Increment (A_I)
prodId	INT	Not Required	None		Untick	PRIMARY	Tick
prodName	VARCHAR	200	None		Untick	--	Untick
prodPicNameSmall	VARCHAR	200	None		Untick	--	Untick
prodPicNameLarge	VARCHAR	200	None		Untick	--	Untick
prodDescripShort	VARCHAR	1000	None		Tick	--	Untick
prodDescripLong	VARCHAR	3000	None		Tick	--	Untick
prodPrice	DECIMAL	8,2	None		Untick	--	Untick
prodQuantity	INT	Not Required	None		Untick	--	Untick

a) 1ST METHOD: Write SQL Code (if you have prior experience in SQL and/or taken the 5COSC020W Database Systems module in semester 1)

- Open a new page on your code editor and save it in your C:\Drive as **homteq_db_tables.sql**.
- Type in a **SQL CREATE statement** to create the **Product** table as specified above. Remember to use the right data types and to create a primary key constraint.
- Paste the code in the SQL window in PHPMysqlAdmin and run it.

b) 2ND METHOD: Use the PHPMysqlAdmin GUI (if you have no prior experience in SQL and not taken the 5COSC020W Database Systems module in semester 1)

- Select your database **w1234567_0** on the left hand-side. Under **Create new table** type in the **name Product** and enter **number of columns 8** and click **Go**.
- Create your **Product** table fields by entering the information specified above.
- In order to ensure referential integrity, make sure that **Storage Engine** is **InnoDB**.
- Click **Save**

Model Answer

To create the Product table, write the following SQL code.

```
CREATE TABLE Product (  
  prodId          INT          AUTO_INCREMENT,  
  prodName        VARCHAR(200) NOT NULL,  
  prodPicNameSmall VARCHAR(200) NOT NULL,  
  prodPicNameLarge VARCHAR(200) NOT NULL,  
  prodDescripShort VARCHAR(1000) ,  
  prodDescripLong  VARCHAR(2000) ,  
  prodPrice        DECIMAL(8,2) NOT NULL DEFAULT '0.00',  
  prodQuantity     INT          NOT NULL DEFAULT '100',  
  CONSTRAINT      p_pid_pk     PRIMARY KEY (prodId)  
) ;
```

TASK 02 | Populate the Product table in the MySQL database

a) **1ST METHOD:** Write SQL Code (if you have prior experience in SQL and/or taken the 5COSC020W Database Systems module in semester 1)

- Access the **homteq_db_tables.sql** on your code editor.
- Type in a **SQL INSERT INTO statement** to insert **4 or 5 new records** in the **Product** table.

```
INSERT INTO Product (      )  
VALUES (      );
```

- Remember to omit the product id from your SQL INSERT INTO code as it will be automatically incremented.
- Go to an online retailer such as John Lewis, PC World or Argos and grab a product name, a short product description and a long product description. Include them in your SQL INSERT INTO code.
- Create an **images** folder in the **homteq** project folder in your C:\Drive.
- From the online retailer, grab a small picture and a large picture of your product.
- Save the actual images in your images folder.
- Include the image names in your SQL INSERT INTO code with the right file extension such as .jpg or .png.
- Include a price and a product quantity available in stock.
- Paste the code in the SQL window in PHPMysqlAdmin and run it.
- Repeat this process 4 times so that you have 4 products in your **Product** table.

b) 2ND METHOD: Use the PHPMysqlAdmin GUI (if you have no prior experience in SQL and not taken the 5COSC020W Database Systems module in semester 1)

- Select the **Product** table on the navigation bar on the left hand-side.
- Click on the *insert* tab.
- Leave the **prodId** value blank as it will be auto incremented
- Go to an online retailer such as John Lewis, PC World or Argos and grab a product name, a short product description and a long product description. Include them in the correct fields of your table insertion GUI.
- Create an **images** folder in the **homteq** project folder in your C:\Drive.
- From the online retailer, grab a small picture and a large picture of your product.
- Save the actual images in your images folder.
- Include the image names in the correct fields of your table insertion GUI with the right file extension such as .jpg or .png.
- Add a price and a product quantity available in stock names in the correct fields of your table insertion GUI.
- After every new product click *Go*.
- Repeat this process 4 times so that you have 4 products in your **Product** table.

NOTE

SQL (Structured Query Language) is a language used to Create, Retrieve, Update and Delete records in a database (CRUD). MySQL is a database Management System.

Model Answer

To insert records in the Product table, write the following SQL code (repeat for multiple products).

```
INSERT INTO Product
(prodName, prodPicNameSmall, prodPicNameLarge, prodDescripShort, prodDescripLong, prodPrice, prodQuantity)
VALUES
('HIVE Active Heating Thermostat', 'hivesmall.jpg', 'hivebig.jpg',
'Hive Active Heating connects you and your heating system via an app on your mobile device to give you convenient
control over operating times from wherever you are.',
'With an all-new wall-mounted thermostat control unit, Hive Active Heating looks great and can reduce the heating
bills in any home. With control from your mobile, laptop or tablet, you\'ll never need to heat an empty home again.
Up to six heating events can be scheduled throughout the day, so you can wake up in comfort, cut the heating while
the house is empty and return to a comfortable home after a long day at work. ',
145.00, 35);
```

TASK 03| Create a db.php database connection file

- 1) Open a new tab in your code editor, type in the following:

db.php

```
<?php
$dbhost = 'phpmyadmin.ecs.westminster.ac.uk';
$dbuser = 'yourWestminsterID';
$dbpass = 'yourMySQLRandomPwd';
$dbname = 'yourWestminsterID_0';

//create a DB connection
$conn = mysqli_connect($dbhost, $dbuser, $dbpass, $dbname);
//if the DB connection fails, display an error message and exit
if (!$conn)
{
    die('Could not connect: ' . mysqli_error($conn));
}
//select the database
mysqli_select_db($conn, $dbname);
?>
```

- 2) Save this file as **db.php** in the **C:\homteq**. When you save it, make sure you select **Save as type: PHP Hypertext Preprocessor file**.
- 3) Go to WinSCP and drag **db.php** from **C:\homteq** (left window pane) and drop it into the **public_html\homteq** folder (right window pane).

TASK 04| Create the INDEX page to display available product names and images (as dynamically retrieved from the database)

- 1) Go to your **C:\homteq** and, in the same folder, copy and paste your **template.php** file.
- 2) Rename the copy to **index.php**.
- 3) Open **index.php** in your code editor and modify it by inserting the sections of code that are highlighted in yellow (code) and green (comments).



Make sure you study and understand the code that you have entered (in yellow) alongside the comments (in green) as you will have to re-use this code in the following tutorial and you will be assessed on your understanding of this code.

index.php

```
<?php
include ("db.php"); //include db.php file to connect to DB

$pageName="Make your home smart"; //create and populate variable called $pageName
echo "<link rel=stylesheet type=text/css href=mystylesheet.css>";
echo "<title>".$pageName."</title>";

echo "<body>";
include ("headfile.html");
echo "<h4>".$pageName."</h4>";

//create a $SQL variable and populate it with a SQL statement that retrieves product details
$SQL="select prodId, prodName, prodPicNameSmall from Product";
//run SQL query for connected DB or exit and display error message
$exeSQL=mysqli_query($conn, $SQL) or die (mysqli_error($conn));

echo "<table style='border: 0px'>";
//create an array of records (2 dimensional variable) called $arrayp.
//populate it with the records retrieved by the SQL query previously executed.
//Iterate through the array i.e while the end of the array has not been reached, run through it
while ($arrayp=mysqli_fetch_array($exeSQL))
{
    echo "<tr>";
    echo "<td style='border: 0px'>";
    //display the small image whose name is contained in the array
    echo "<img src=images/'".$arrayp['prodPicNameSmall']."' height=200 width=200>";
    echo "</td>";
    echo "<td style='border: 0px'>";
    echo "<p><h5>".$arrayp['prodName']."'</h5>"; //display product name as contained in the array
    echo "</td>";
    echo "</tr>";
}
echo "</table>";
include ("footfile.html");
echo "</body>";
?>
```

- 4) Go to WinSCP and drag index.php from C:\homteq (left window pane) and drop it into the public_html\homteq folder (right window pane). Select 'Replace all'.

- 5) Go to WinSCP and also drag the **images** folder from **C:\homteq** (left window pane) and drop it into the **public_html\homteq** folder (right window pane). Select 'Replace all'.
- 6) Open a Web browser and type in <https://yourWestminsterID.users.ecs.westminster.ac.uk/homteq/index.php> with YOUR Westminster identifier.

You should be able to see a display similar to this:

index.php

make your home smart

https://rouberf.users.ecs.westminster.ac.uk/homteq/index.php 90%

homteq
your home for home tech

London, Sunday 10 January 2021 14:03:53

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MAKE YOUR HOME SMART



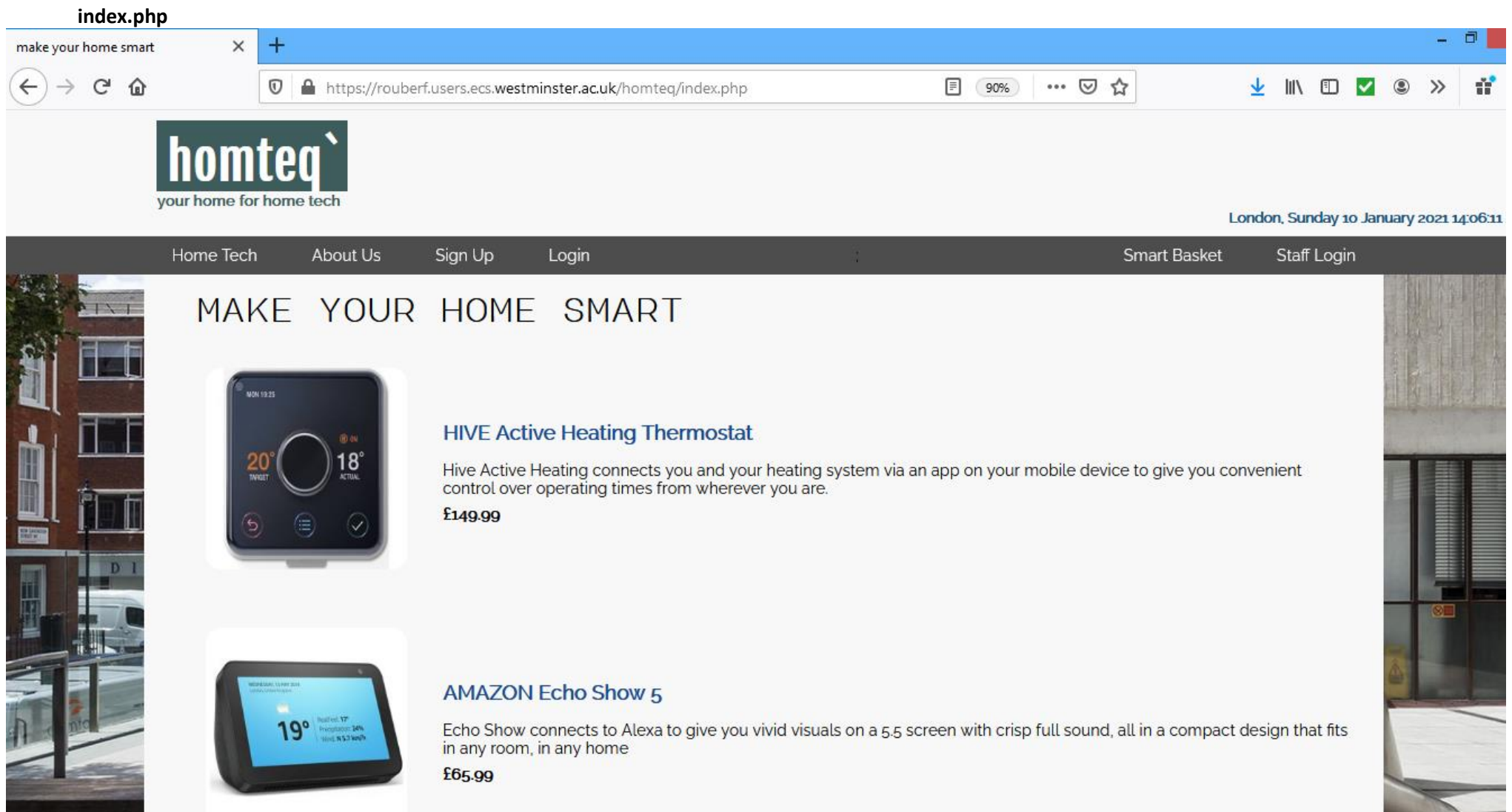
HIVE Active Heating Thermostat



AMAZON Echo Show 5

TASK 05 | Modify the INDEX page to also display available product descriptions and prices

- 1) Modify the code on **index.php** so that to also display, underneath the name of the product, a short product description and the product price (*hint: also edit SQL*)
- 2) Upload the modified **index.php** file using WinSCP and refresh your browser. You should be able to see a display similar to this:



On the left hand-side, the display should show the small image.

On the right-hand side, the display should show the product name, the short description and the product price.

Model Answer

To display the short description and the price of the product, you first need to modify the SQL query as follows:

```
$SQL="select prodId, prodName, prodPicNameSmall, prodDescripShort, prodPrice from Product";
```

Then you need to add these additional PHP lines under the display of the product name.

```
echo "<p>".$arrayp['prodDescripShort'];  
echo "<p>&pound".$arrayp['prodPrice'];
```

TASK 06| Modify the INDEX page to make the images into clickable anchors that link to the PRODUCT page

- 1) Access the **index.php** file and locate the line of PHP code that displays the image using the **** tag.
- 2) Right before the image line, insert the following PHP code.
This creates an anchor tag **<a>** which specifies the HTML reference **prodbuy.php** to which the user is going to get directed when the user clicks on the anchor. It also specifies that the product id gets appended to the URL and passed to **prodbuy.php** using the GET method. To do this, it uses a query string i.e a name/value pair which here is **prod_id** for the name and **\$arrayp['prodId']** for the value.

```
//make the image into an anchor to prodbuy.php and pass the product id by URL (the id from the array)  
echo "<a href=prodbuy.php?u_prod_id=".$arrayp['prodId']. ">";
```

- 3) Right after the image line, close the anchor tag **** as follows.

```
//close the anchor  
echo "</a>";
```

- 4) Upload the modified **index.php** file using WinSCP and refresh your browser.

You should be able to see a display identical to the previous one, but you should be able to hover on each product name and image and a link to the following URL should be activated with a reference to the selected product id, as illustrated on the display below.

https://yourWestminsterID.users.ecs.westminster.ac.uk/homteq/prodbuy.php?prod_id=... (with your Westminster identifier).

index.php

make your home smart

https://rouberf.users.ecs.westminster.ac.uk/homteq/index.php

90%

London, Sunday 10 January 2021 14:06:11

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MAKE YOUR HOME SMART



HIVE Active Heating Thermostat

Hive Active Heating connects you and your heating system via an app on your mobile device to give you convenient control over operating times from wherever you are.

£149.99



AMAZON Echo Show 5

Echo Show connects to Alexa to give you vivid visuals on a 5.5 screen with crisp full sound, all in a compact design that fits in any room, in any home

£65.99

https://rouberf.users.ecs.westminster.ac.uk/homteq/prodbuy.php?prod_id=2

Check that when you hover the name of the product or the image with the mouse, you can see the URL appearing. It should pick up the id number of the selected product and show it as a reference right at the end of the URL (see immediately above).